Summer 1981

Problem 1 (Su81). Let $y(h) = 1 - 2\sin^2(2\pi h)$ and $f(y) = \dfrac{2}{1+\sqrt{1-y}}$. Justify the statement $f\,(y(h)) = 2 - 4\sqrt{2}\pi\,|h| + O\left(h^2\right)$ as $h \to 0$.

Problem 2 (Su81). Let G be a finite group, and let φ be an automorphism of G which leaves fixed only the identity element of G.

1. Show that every element of G may be written in the form $g^{-1}\varphi(g)$.
2. If φ has order 2 show that φ is given by the formula $g \mapsto g^{-1}$ and that G is an abelian group whose order is odd.

Problem 3 (Su81). Prove or disprove: The set $\mathbb{Q}$ of rational numbers is the intersection of a countable family of open subsets of $\mathbb{R}$.

Problem 4 (Su81). Let $f : \mathbb{R} \to \mathbb{R}$ be continuous, $\displaystyle\int_{-\infty}^{\infty} |f(x)|\,\mathrm{d}x < \infty$. Show that there is a sequence (x_n) such that $x_n \to \infty$, $x_n f\,(x_n) \to 0$, and $x_n f\,(-x_n) \to 0$ as $n \to \infty$.

Problem 5 (Su81). Let S denote the vector space of real $n \times n$ skew-symmetric matrices. For a nonsingular matrix A, compute the determinant of the linear map $T_A : S \to S$, $T_A(X) = AXA^{\mathsf{T}}$.

Problem 6 (Su81, Sp14). Let $\mathrm{SO}_3(\mathbb{R})$ denote the group of orthogonal transformations of $\mathbb{R}^3$ of determinant 1. Let $Q \subset \mathrm{SO}_3(\mathbb{R})$ be the subset of symmetric transformations $\neq I$. Let P^2 denote the space of lines through the origin in $\mathbb{R}^3$.

1. Show that P^2 and $\mathrm{SO}_3(\mathbb{R})$ are compact metric spaces, in their usual topologies.
2. Show that P^2 and Q are homeomorphic.

Problem 7 (Su81, Fa24). Compute

$$\frac{1}{2\pi i}\int_C \frac{\mathrm{d}z}{\sin 1/z}$$

where C is the circle $|z| = \dfrac{1}{5}$ positively oriented.

Problem 8 (Su81, Su82). Show that $x^{10}+x^9+x^8+\cdots+x+1$ is irreducible over $\mathbb{Q}$. How about $x^{11}+x^{10}+\cdots+x+1$?

Problem 9 (Su81). Let $f : \mathbb{R} \to \mathbb{R}$ be the function of period 2π such that $f(x) = x^3$ for $-\pi \leqslant x < \pi$.

1. Prove that the Fourier series for f has the form $\sum_1^\infty b_n \sin nx$ and write an integral formula for b_n (do not evaluate it).

2. Prove that the Fourier series converges for all x.

3. Prove $\sum_{n=1}^{\infty} b_n^2 = \frac{2\pi^6}{7}$

Problem 10 (Fa79, Su81, Fa19). Let V be the vector space of sequences (a_n) of complex numbers. The shift operator $S : V \to V$ is defined by

$$S\left((a_1, a_2, a_3, \ldots)\right) = (a_2, a_3, a_4, \ldots)$$

1. Find the eigenvectors of S.

2. Show that the subspace W consisting of the sequences (x_n) with $x_{n+2} = x_{n+1}+x_n$ is a two-dimensional, S-invariant subspace of V and exhibit an explicit basis for W.

3. Find an explicit formula for the n-th Fibonacci number F_n, where $F_2 = F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$ for $n \geqslant 1$.

Note: See also **??**.

Problem 11 (Su81). Show that the equation

$$x\left(1 + \log\left(\frac{1}{\varepsilon\sqrt{x}}\right)\right) = 1 \quad x > 0 \quad \varepsilon > 0$$

has, for each sufficiently small $\varepsilon > 0$, exactly two solutions. Let $x(\varepsilon)$ be the smaller one. Show that

1. $x(\varepsilon) \to 0$ as $\varepsilon \to 0+$;

yet for any $s > 0$,

2. $\varepsilon^{-s}x(\varepsilon) \to \infty$ as $\varepsilon \to 0+$.

Problem 12 (Su81, Sp93). Show that no commutative ring with identity has additive group isomorphic to $\mathbb{Q}/\mathbb{Z}$.

Problem 13 (Su81). Let G be an additive group, and $u, v : G \to G$ homomorphisms. Show that the map $f : G \to G$, $f(x) = x - v(u(x))$ is surjective if the map $h : G \to G$, $h(x) = x - u(v(x))$ is surjective.

Problem 14 (Su81). Let $I \subset \mathbb{R}$ be the open interval from 0 to 1. Let $f : I \to \mathbb{C}$ be $\mathcal{C}^1$, i.e., the real and imaginary parts are continuously differentiable. Suppose that $f(t) \to 0$, $f'(t) \to C \neq 0$ as $t \to 0+$. Show that the function $g(t) = |f(t)|$ is $\mathcal{C}^1$ for sufficiently small $t > 0$ and that $\lim_{t\to 0+} g'(t)$ exists, and evaluate the limit.

Problem 15 (Su81, Su82). Let V be a finite dimensional vector space over the rationals $\mathbb{Q}$ and let M be an automorphism of V such that M fixes no nonzero vector in V. Suppose that M^p is the identity map on V, where p is a prime number. Show that the dimension of V is divisible by $p - 1$.

Problem 16 (Su81). Let $\{f_n\}$ be a sequence of continuous functions defined from $[0, 1] \to \mathbb{R}$ such that

$$\int_0^1 (f_n(x) - f_m(x))^2 \, dx \to 0 \quad n, m \to \infty$$

Let $K : [0, 1] \times [0, 1] \to \mathbb{R}$ be continuous. Define $g_n : [0, 1] \to \mathbb{R}$ by

$$g_n(x) = \int_0^1 K(x, y) f_n(y) \, dy$$

Prove that the sequence $\{g_n\}$ converges uniformly.

Problem 17 (Fa79, Su81, Sp14). Suppose f and g are entire functions with $|f(z)| \leqslant |g(z)|$ for all z. Prove that $f(z) = cg(z)$ for some constant c.

Problem 18 (Su81, Fa16). If two rational matrices are similar over $\mathbb{C}$, then they are similar over $\mathbb{Q}$.

Problem 19 (Su81). Prove that the number of roots of the equation

$$z^{2n} + \alpha^2 z^{2n-1} + \beta^2 = 0$$

with n a natural number, α and β real nonzero; that have positive real part is n if n is even, and $n - 1$ if n is odd.

Problem 20 (Fa79, Su81, Fa92). Let $y = y(x)$ be a solution of the differential equation $y'' = -|y|$ with $-\infty < x < \infty$, $y(0) = 1$ and $y'(0) = 0$.

1. Show that y is an even function.

2. Show that y has exactly one zero on the positive real axis.